Installation and Demonstration — Windows PowerShell

From repository root:

python -m venv .venv

.\.venv\Scripts\Activate.ps1

python -m pip install -r requirements.txt

Copy-Item .env.example .env

Edit .env locally. Start with DEFAULT_LLM_PROVIDER=mock.

python -m pytest -q

Terminal 1 — API:

python -m uvicorn backend.main:app --reload

Open http://127.0.0.1:8000/health and http://127.0.0.1:8000/docs.

Terminal 2 — UI:

.\.venv\Scripts\Activate.ps1

python -m streamlit run frontend/app.py

Open http://localhost:8501. Upload data/samples/AUTOSAR_HLD_Sample_BrakingSystem.pdf; inspect metrics, SWC/Port/Interface tables; ask a document-answerable question, open verified citations; ask an absent-fact question and inspect whether it refuses.

Real Gemini: store GEMINI_API_KEY privately in .env; select DEFAULT_LLM_PROVIDER=gemini and an available model; restart both services. Capture the provider label, factual answer and source evidence without leaking secrets.

MockLLM is an integration test double, not a measure of natural answer quality. Real LLM results and negative grounding require human inspection.